

Golden Section Search on Lognormal Density — Algorithm, Code & Output

Algorithm

Same golden-section engine as before ($\tau=(\sqrt{5}-1)/2\approx 0.618$), now minimizing $f(x)=-\text{lognormal density}$ — sign flip turns “find the mode (max)” into “minimize -density”, no algorithm change needed.

- 1) Place $x_1=b-\tau(b-a)$, $x_2=a+\tau(b-a)$; evaluate f at both (domain $x>0$, so start $a=0.01$ not 0).
- 2) Compare: $f(x_1)>f(x_2) \rightarrow$ drop left, $a \leftarrow x_1$; else drop right, $b \leftarrow x_2$. Reuse old point + f -value $\rightarrow$ 1 new eval/iteration.
- 3) Repeat while $(b-a) > \text{tol}$; return midpoint $(a+b)/2$ as $x_{\min}$. Wider start $[0.01,10]$ vs $[1,5]$ $\rightarrow$ more iterations to hit same tol .

Code (short)

```
f <- function(x) -(1/x*sqrt(2*pi))*exp(-0.5*(log(x)*log(x)))

golden_search <- function(f,a,b,tol=1e-4){
  tau <- (sqrt(5)-1)/2
  hist <- data.frame(iter=0,a=a,b=b)
  x1<-b-tau*(b-a); x2<-a+tau*(b-a)
  f1<-f(x1); f2<-f(x2); iter<-0
  while((b-a) > tol){
    iter<-iter+1
    if(f1>f2){ a<-x1;x1<-x2;f1<-f2
      x2<-a+tau*(b-a); f2<-f(x2) }
    else{ b<-x2;x2<-x1;f2<-f1
      x1<-b-tau*(b-a); f1<-f(x1) }
    hist<-rbind(hist,data.frame(iter,a,b))
  }
  list(xmin=(a+b)/2, interval=c(a,b), history=hist)
}
result <- golden_search(f, a=0.01, b=10, tol=1e-4)
```

Output

Run on $f(x) = -(\text{lognormal density})$, $[0.01,10]$, $\text{tol}=1e-4 \rightarrow 24$ iterations.

$x_{\min} \approx 0.367858$ $\text{interval} \approx [0.367810, 0.367906]$

Matches closed-form lognormal mode $e^{-1}\approx 0.367879$ exactly — confirms correctness. Plot: red/green lines (a,b per step) funnel onto the single sharp dip near $x\approx 0.37$; note plot window $[0,6]$ doesn't show early $b=10$ line, though search itself used full $[0.01,10]$.